

The original real conductor: every resonance direction and the complete pole transition

Split-Zero programme: coordinate calculation

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Result SZ-20260921-006 supplies the complete RPD1–31 proof, including RPD12a. The complete earlier certified-root and four-label proofs accompany it in `REAL_PAIR_005.tex`, its readable PDF, and `FABLE_TO_ORIGINAL_CONDUCTOR.tex`. The original-source citations below identify the precise public providers. Every claim concerns the specified geometric family.

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1 The complete real directional geometry at the certified pair

This continuation uses the exact point proved in RPZ1–19 [5], not a numerically selected substitute. It calculates the original two real parameters simultaneously. The original family is [1, PCL1–10]; its pinned complete proof specifies the period conjugation. The unchanged norms and weighted matrix are [6, WCF1–6], with their pinned complete proof. RPCJ1–23 and RPAC1–15 [3, 2] prove the preceding fixed-direction formulas; the calculation below proves their full directional and curved-path extension. Those complete sources and the RPZ certificate accompany this derivation. The complete four-label return is RPE1–26 [4]. The four ES labels, original quartet, and fixed amplification parameter are retained. No geometric direction is identified with an arithmetic zeta direction.

1.1 Original functions and the exact derivative isomorphism

Write $p = (z, x)$ and $p_* = (z_*, x_*)$, the unique real zero in RPZ4's specified rational rectangle. Retain $u = 1/z$, $a = \alpha(1 + ix)$ with its original nonzero real α , $\delta = 1/4$, $\gamma = 3$, $\beta = 143/8$, $\eta = 21025/256$, $w = (\delta + i\gamma, \delta - i\gamma, -\delta + i\gamma, -\delta - i\gamma)$, $\rho_a^{\text{prim}} = 1/2 + w_a$, and $V_{ar} = w_a^r$. The superscript distinguishes these four primary exponents from the Gamma weights below. With $J_d = \text{diag}(1, -1, -1, 1)$, $W(x) = I + ixJ_d$, $\zeta = e^{2\pi i/5}$, the full original functions are

$$\begin{aligned}
\mathcal{R}(z) &= \sum_{n \geq 0} R_n z^n, \\
(R_n)_{ar} &= \sum_{2h+4k=5n+r+1-a} \frac{(\beta/3)^h \eta^k (-5)^{h+k-n} (a/5)_{h+k-n}}{h!k!}, \\
M_j(p) &= W(x)^{-1} V \mathcal{R}(z)^{-1} D_j \mathcal{R}(z) V^{-1} W(x), \\
D_j &= \text{diag}(\zeta^{-j}, \dots, \zeta^{-4j}), \\
f_j(\xi, p) &= Q_0(M_j(p)(e^{\rho_a^{\text{prim}} \xi})_a), \\
Q_0(y) &= y_1 y_4 - y_2 y_3, \\
G(\xi, p) &= f_1(\xi, p) f_2(\xi, p), \\
E_A(\xi, p) &= \prod_{j=1}^4 f_j(\xi, p).
\end{aligned} \tag{RPD1}$$

Here $1 \leq a \leq 4$, $0 \leq r \leq 3$, and the sum has $h, k \geq 0$; empty sums are zero. These are the same coefficients as RPZ2, with summation letters changed to avoid the parameter p . RPZ5–9 prove convergence and all differentiation used below and prove invertibility of the matrices on a neighborhood of p_* . The common α cancels only in its matrix conjugation. Conjugating the original matrices exactly as in RPCJ2, now with both real parameters varying, gives

$$f_{5-j}(\xi, p) = -\overline{f_j(\bar{\xi}, p)}, \quad E_A(\xi, p) = G(\xi, p) \overline{G(\bar{\xi}, p)} \quad (p \in \mathbb{R}^2). \tag{RPD2}$$

Indeed the real series $\mathcal{R}$, the conjugate-paired V , and $\overline{W(x)} = J_c W(x) J_c$ obey the identical products for every real x ; J_c exchanges indices 1, 2 and 3, 4, and $Q_0(J_c y) = -Q_0(y)$. Thus RPD2 retains both minus signs, and $E_A(\xi, p) = |G(\xi, p)|^2$ for real ξ, p . Write $\mu_n(p) = \partial_\xi^n E_A(0, p)$. All path limits below are real; fractional powers use $\tau > 0$ explicitly.

All derivatives in the following definitions are at $(0, p_*)$:

$$\begin{aligned}
b_* &= f_2 \neq 0, \quad f_1 = 0, \quad B = G_\xi = b_* f_{1,\xi} \neq 0, \\
A_z &= G_z = b_* f_{1,z}, \quad A_x = G_x = b_* f_{1,x}, \\
\Lambda : \mathbb{R}^2 &\longrightarrow \mathbb{C}, \quad \Lambda(v) = A_z v_z + A_x v_x, \quad \mathcal{J} = \begin{pmatrix} \Re A_z & \Re A_x \\ \Im A_z & \Im A_x \end{pmatrix}.
\end{aligned} \tag{RPD3}$$

Here $\mathbb{C}$ is a two-dimensional real vector space. RPZ10–13 give an invertible exact real P and $\|I - PJ\|_r < 1$, where $J = D_p(\Re f_1, \Im f_1)|_{p_*}$. The geometric series in $I - PJ$ inverts PJ , so J is invertible. Multiplication by b_* has real matrix $\begin{pmatrix} \Re b_* & -\Im b_* \\ \Im b_* & \Re b_* \end{pmatrix}$ and determinant $|b_*|^2 > 0$. The product rule and $f_1 = 0$ give $\mathcal{J} = \text{Mat}_{\mathbb{R}}(b_*)J$. Consequently

$$\Delta = \det \mathcal{J} = \Im(\overline{A_z} A_x) = |b_*|^2 \det J \neq 0, \quad \Lambda^{-1}(q) = \frac{1}{\Delta} \begin{pmatrix} \Im(A_x \bar{q}) \\ \Im(\overline{A_z} q) \end{pmatrix}. \quad (\text{RPD4})$$

Multiplying this vector by the two rows of $\mathcal{J}$ proves the inverse formula, including both signs. This is an exact morphism, not an identification of two coordinate presentations by assertion. It also proves $A_v := \Lambda(v) \neq 0$ for every real $v \neq 0$. For reference its orientation is negative: the real continuous matrices $I - t(I - PJ)$ are invertible for $0 \leq t \leq 1$, so $\det(PJ) > 0$, whereas RPZ11 proves $\det P < 0$.

1.2 The full quadratic form and the real spectral zero curve

For $h = (h_z, h_x)$ the original zeroth moment has the expansion

$$\begin{aligned} \mu_0(p_* + h) &= h^\top \mathcal{Q} h + O(\|h\|^3), \\ \mathcal{Q} &= \mathcal{J}^\top \mathcal{J} = \begin{pmatrix} |A_z|^2 & \Re(A_z \overline{A_x}) \\ \Re(\overline{A_z} A_x) & |A_x|^2 \end{pmatrix}, \\ \det \mathcal{Q} &= \Delta^2 > 0, \quad D_p^2 \mu_0(p_*) = 2\mathcal{Q}. \end{aligned} \quad (\text{RPD5})$$

This follows by Taylor expanding $G(0, p_* + h) = \Lambda(h) + O(\|h\|^2)$ and multiplying by its conjugate. Thus the quadratic form is positive definite in the unchanged two original parameters. On RPZ4's rectangle, f_2 never vanishes and f_1 has its unique zero at p_* ; hence $\mu_0 > 0$ at every other real point of that rectangle. The joint real expansion, retaining the spectral coordinate, is

$$E_A(\xi, p_* + h) = |B\xi + \Lambda(h)|^2 + O((|\xi| + \|h\|)^3). \quad (\text{RPD6})$$

Its real quadratic form has rank two and its exact null line is $\{(\xi, -\xi\Lambda^{-1}(B)) : \xi \in \mathbb{R}\}$. The actual zero set is a smooth curve, not this tangent line alone.

Here is a construction of that curve without substituting a formal tangent for the actual equation. Let $\mathcal{H}_\xi(h) = h - \Lambda^{-1}G(\xi, p_* + h)$, with complex outputs interpreted as real vectors. At $(\xi, h) = (0, 0)$ its derivative in h is zero. Choose a closed sufficiently small real ball on which its derivative norm is at most $1/2$, and then restrict $|\xi|$ so $\|\mathcal{H}_\xi(0)\|$ is at most half the radius. Continuity of the complete derivatives in RPD1 guarantees these two strict inequalities after reducing the radius and interval. The map preserves the ball and contracts it. Iteration from zero converges by the geometric difference estimate in RPZ14 and gives its unique fixed point $h(\xi)$. Each fixed point is precisely a zero of G . The same contraction estimate gives continuous, locally Lipschitz dependence on ξ ; Taylor subtraction of the two zero equations then gives differentiability with the invertible parameter derivative. Repeated differentiation gives every derivative because the full functions are smooth. Every real zero nearby is this fixed point. Writing $p(\xi) = p_* + h(\xi)$, its first two derivatives are exactly

$$\begin{aligned} v_0 &= p'(0) = -\Lambda^{-1}(B), \\ p''(0) &= -\Lambda^{-1}(F + 2\mathcal{H}(v_0) + \mathcal{C}(v_0, v_0)), \\ F &= G_{\xi\xi}, \quad \mathcal{H}(v) = G_{\xi z}v_z + G_{\xi x}v_x, \quad \mathcal{C}(v, w) = \sum_{i,j \in \{z,x\}} G_{ij}v_iw_j. \end{aligned} \quad (\text{RPD7})$$

Differentiating $G(\xi, p(\xi)) = 0$ twice proves the formula. Because $E_A = |G|^2$, its zero set on this real neighborhood is exactly the same curve.

1.3 Full original jets along each nonzero direction

Fix an arbitrary nonzero $v \in \mathbb{R}^2$ and use the literal path $p = p_* + \tau v$. Put $A_v = \Lambda(v)$, $C_v = \mathcal{C}(v, v)$ and $H_v = \mathcal{H}(v)$. Every derivative here is determined by the original factors:

$$\begin{aligned} C_v &= b_* \partial_v^2 f_1 + 2(\partial_v f_1)(\partial_v f_2), \\ H_v &= b_* \partial_v \partial_\xi f_1 + (\partial_v f_1) \partial_\xi f_2 + (\partial_\xi f_1) \partial_v f_2, \\ F &= b_* \partial_\xi^2 f_1 + 2(\partial_\xi f_1)(\partial_\xi f_2), \\ \partial_v^m \mu_n(p_*) &= \sum_{r=0}^n \sum_{j=0}^m \binom{n}{r} \binom{m}{j} (\partial_\xi^r \partial_v^j G) \overline{\partial_\xi^{n-r} \partial_v^{m-j} G}. \end{aligned} \tag{RPD8}$$

The first three identities are the product rule with $f_1 = 0$; the last differentiates RPD2 and therefore includes every moment. Define the exact directional constants

$$\begin{aligned} c_v &= |A_v|^2 > 0, & R_v &= 2\Re(A_v \bar{B}), & b &= |B|^2 > 0, \\ d_v &= -iR_v, & e &= -b, & c_{3,v} &= \Re(C_v \bar{A}_v), \\ d_{2,v} &= -i\{\Re(\bar{B} C_v) + 2\Re(H_v \bar{A}_v) - 4c_v\}, \\ e_{1,v} &= -\Re(F \bar{A}_v) - 2\Re(H_v \bar{B}) + 4R_v, \\ f &= i\{\Re(F \bar{B}) - 4b\}. \end{aligned} \tag{RPD9}$$

These constants are not independent choices of jets. For the complete original symbol $g(t, p) = e^{-4\omega(t)} E_A(\omega(t), p)$, $\omega(t) = -i \arctan t$, coefficient differentiation gives

$$\begin{aligned} g_0(p_* + \tau v) &= c_v \tau^2 + c_{3,v} \tau^3 + O(\tau^4), \\ g_1(p_* + \tau v) &= d_v \tau + d_{2,v} \tau^2 + O(\tau^3), \\ g_2(p_* + \tau v) &= e + e_{1,v} \tau + O(\tau^2), & g_3(p_* + \tau v) &= f + O(\tau). \end{aligned} \tag{RPD10}$$

For clarity the identities before taking these jets are $g_0 = \mu_0$, $g_1 = -i(\mu_1 - 4\mu_0)$, $g_2 = -\mu_2/2 + 4\mu_1 - 8\mu_0$, and $g_3 = i(\mu_3/6 - 2\mu_2 + 25\mu_1/3 - 12\mu_0)$. They follow by expanding the exact ω and exponential through the stated degree; RPD8 supplies their complete parameter jets.

1.4 Every cutoff and its full metric constants

Retain $s \in \{1, a_{\text{amp}}\}$ and the original centers $a_{\text{amp}}/2$ and $a_{\text{amp}}/2 - 4$. The physical measure and weights, with no removal of mass or center, are

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, & M_s &= (2\pi)^{s/2}, \\ \rho_j &= \sqrt{j!/(s/2)_j}, & N &= D + 1, & B_{D,s}(p) &= \text{diag}(\rho)^{-1} T_D(g(\cdot, p)) \text{diag}(\rho), \\ T_D(g)_{ij} &= g_{j-i} \ (j \geq i), & T_D(g)_{ij} &= 0 \ (j < i). \end{aligned} \tag{RPD11}$$

These are the original orthonormal frames of WCF2–6. Its two mass factors cancel only in the displayed matrix representation. In particular the fixed old map has domain and codomain

$$(\mathcal{P}_D, \|\cdot\|_{s, a_{\text{amp}}/2}) \quad \text{and} \quad (\mathcal{P}_D, \|\cdot\|_{s, a_{\text{amp}}/2-4}),$$

where $\mathcal{P}_D = \mathbb{C}[S]_{\leq D}$ and $\|P\|_{s,c}^2 = \int_{\mathbb{R}} |P(c + iy)|^2 dm_s(y)$. At p_* the fixed old matrix has rank $N - 2$ for $D \geq 1$. For $D \geq 2$ its full nonzero block $C_{D,s}$ has rows $0, \dots, D - 2$ and columns $2, \dots, D$, with entries

$$(C_{D,s})_{ij} = g_{j+2-i}(p_*) \rho_{j+2}/\rho_i \quad (j \geq i), \quad (C_{D,s})_{ij} = 0 \quad (j < i). \tag{RPD12}$$

The diagonal $e\rho_{i+2}/\rho_i$ never vanishes. All higher coefficients $g_j(p_*)$ remain in this block. For $D = 1$ it is empty, with zeroth exterior norm and determinant equal to one. The actual order-two quotient at the point retains a different domain, and its exact original matrix is

$$\begin{aligned} \mathcal{A}_{D,s} : \mathcal{P}_{D+2}/\mathcal{P}_1 &\longrightarrow \mathcal{P}_D, & (\mathcal{A}_{D,s})_{ij} &= g_{j-i+2}(p_*)\rho_{j+2}/\rho_i \quad (j \geq i), \\ (\mathcal{A}_{D,s})_{ij} &= 0 \quad (j < i), & (\mathcal{A}_{D,s})_{ii} &= e\rho_{i+2}/\rho_i \neq 0. \end{aligned} \quad (\text{RPD12a})$$

The source has the original quotient norm at center $a_{\text{amp}}/2$, and the target has its original norm at center $a_{\text{amp}}/2 - 4$. WCF4–6 applied to the complete symbol gives RPD12a: the original polynomial map sends S^n to $\sum_j \binom{n}{j} \mu_j S^{n-j}$, its first nonzero moment is $\mu_2 = 2|B|^2$, and its quotient symbol is $t^{-2}g(t, p_*)$. The nonzero diagonal proves this quotient map invertible. Changing its domain does not remove the poles of the fixed old map just specified in RPD11.

For $n \geq 0$ define finite expressions

$$\begin{aligned} H_n(v) &= \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^{n-j} \binom{n-j}{j} \frac{d_v^{n-2j} e^j}{c_v^{n-j+1}}, \quad H_{-1} = H_{-2} = 0, \\ K_n(v) &= -[y^n] \frac{c_{3,v} + d_{2,v}y + e_{1,v}y^2 + fy^3}{(c_v + d_vy + ey^2)^2} \\ &= - \sum_{j=0}^{\min(3,n)} a_{j,v} \sum_{k=0}^{n-j} H_k(v) H_{n-j-k}(v), \quad (a_{0,v}, a_{1,v}, a_{2,v}, a_{3,v}) = (c_{3,v}, d_{2,v}, e_{1,v}, f). \end{aligned} \quad (\text{RPD13})$$

All higher original coefficients remain in g . The exact inverse entries, for $0 \leq n \leq D$, satisfy

$$[t^n]g(t, p_* + \tau v)^{-1} = \tau^{-n-2} \{H_n(v) + \tau K_n(v) + O(\tau^2)\}. \quad (\text{RPD14})$$

To prove this, the exact coefficient substitution $t = \tau y$ gives $g(\tau y, p_* + \tau v) = \tau^2 q_v(y) + \tau^3 a_v(y) + O(\tau^4)$, where $q_v = c_v + d_v y + e y^2$ and a_v is the numerator in RPD13. Reciprocation gives $q_v^{-1} - \tau a_v q_v^{-2} + O(\tau^2)$ after the exact factor τ^2 is retained. Coefficient extraction proves RPD14 and geometric expansion proves RPD13. This is a coefficient identity for each finite cutoff, not deletion of higher moments.

Put $\lambda_v = |B|/|A_v|$ and $\theta_v = \Re(A_v \bar{B})/(|A_v||B|)$. Multiplication by q_v gives $c_v H_n + d_v H_{n-1} + e H_{n-2} = \mathbf{1}_{n=0}$. Defining $U_0 = 1$, $U_1 = 2X$, $U_n = 2XU_{n-1} - U_{n-2}$, induction and the sine addition identity prove

$$\begin{aligned} H_n(v) &= \frac{(i\lambda_v)^n}{c_v} U_n(\theta_v), \quad H_{n-1}(v)^2 - H_n(v)H_{n-2}(v) = \frac{e^{n-1}}{c_v^{n+1}} \quad (n \geq 1), \\ H_D(v) = 0 &\iff \theta_v = \cos \frac{k\pi}{D+1} \quad \text{for some } 1 \leq k \leq D. \end{aligned} \quad (\text{RPD15})$$

The first identity holds for every $n \geq 0$. For the second identity the initial value is c_v^{-2} and the recurrence multiplies it by e/c_v at each step. For the last, $U_D(\cos \phi) = \sin((D+1)\phi)/\sin \phi$; the D displayed roots are distinct, and U_D has degree D and leading coefficient 2^D . These arguments prove all degrees, with no finite-degree test standing in for the proof.

For $2 \leq r \leq N$ define the positive constants

$$L_{r,D,s}(v) = \frac{\|\bigwedge^{N-r} C_{D,s}\|}{c_v^N}, \quad L_{2,D,s}(v) = \frac{|e|^{D-1} \rho_{D-1} \rho_D}{c_v^{D+1} \rho_0 \rho_1}. \quad (\text{RPD16})$$

The second equality multiplies the complete block diagonal and telescopes its weights. The exact singular-value identity $\|\bigwedge^r B^{-1}\| = \|\bigwedge^{N-r} B\|/|\det B|$ follows by multiplying the corresponding singular values. Here $\det B = g_0^N$, so continuity and RPD10 prove

$$\lim_{\tau \rightarrow 0} |\tau|^{2N} \left\| \bigwedge^r B_{D,s}(p_* + \tau v)^{-1} \right\| = L_{r,D,s}(v) \quad (2 \leq r \leq N). \quad (\text{RPD17})$$

When $H_D(v) \neq 0$, the sole entry at pole $D + 2$ is the top right. The two divergent inverse singular values, in decreasing order, have poles and constants

$$(D + 2, D), \quad \left(a_{D,s}(v), \frac{L_{2,D,s}(v)}{a_{D,s}(v)} \right), \quad a_{D,s}(v) = |H_D(v)|\rho_D/\rho_0. \quad (\text{RPD18})$$

Indeed RPD14 gives the first limit; division of RPD17 at $r = 2$ gives the second. When $H_D(v) = 0$, RPD15 forces $H_{D-1}(v) \neq 0$. The full leading block of $\tau^{D+1}B_{D,s}^{-1}$ is

$$N_{D,s}(v) = \begin{pmatrix} H_{D-1}(v)\rho_{D-1}/\rho_0 & K_D(v)\rho_D/\rho_0 \\ 0 & H_{D-1}(v)\rho_D/\rho_1 \end{pmatrix}. \quad (\text{RPD19})$$

It occupies rows $0, 1$ and columns $D - 1, D$, and is invertible. Every remaining entry tends to zero. Therefore both divergent inverse singular values have pole $D + 1$, with constants $\sigma_+(N_{D,s})$ and $\sigma_-(N_{D,s})$. For entries $(u, w; 0, v_1)$ these constants equal $\{(S \pm \sqrt{S^2 - 4|uv_1|^2})/2\}^{1/2}$, $S = |u|^2 + |w|^2 + |v_1|^2$, retaining the complete upper-right term. In both cases all other inverse singular values tend to reciprocals of the positive singular values of $C_{D,s}$ in reverse order. For $D = 1$ the block RPD19 is the full matrix; for $D = 0$ the only inverse pole is two with constant c_v^{-1} .

1.5 All resonance rays in the original coordinates

For $D \geq 1$, $1 \leq k \leq D$, $\varepsilon \in \{1, -1\}$ and $r > 0$ put $\phi_k = k\pi/(D + 1)$ and define the exact real vectors

$$\begin{aligned} v_{D,k,\varepsilon}(r) &= \Lambda^{-1}(rBe^{i\varepsilon\phi_k}) \\ &= \frac{r}{\Delta} \begin{pmatrix} \Im(A_x \bar{B}e^{-i\varepsilon\phi_k}) \\ \Im(\bar{A}_z Be^{i\varepsilon\phi_k}) \end{pmatrix}. \end{aligned} \quad (\text{RPD20})$$

Their actual values satisfy $A_v = rBe^{i\varepsilon\phi_k}$, $c_v = r^2|B|^2$, $R_v = 2r|B|^2 \cos \phi_k$ and $\lambda_v = 1/r$. No magnitude is discarded: r remains arbitrary and all its powers remain in RPD9–19. They give every resonant direction. In fact for any nonzero v the number A_v/B is nonzero, and the equality $\Re(A_v/B)/|A_v/B| = \cos \phi_k$ forces $A_v/B = re^{i\phi_k}$ or $re^{-i\phi_k}$ for its actual $r > 0$. Applying the inverse in RPD4 proves exhaustiveness.

For positive rays there are exactly $2D$ distinct choices, since their image arguments $\pm\phi_k$ are distinct modulo 2π . The exact duplication upon forgetting orientation is

$$v_{D,D+1-k,-\varepsilon}(r) = -v_{D,k,\varepsilon}(r). \quad (\text{RPD21})$$

Thus there are exactly D unoriented real lines. There is one line with $\theta = 0$ precisely when D is odd; it corresponds to $k = (D + 1)/2$. All remaining values of θ are paired with $-\theta$. Under any real scaling $\ell \neq 0$, direct substitution in RPD9 and RPD13 gives

$$\begin{aligned} \theta_{\ell v} &= \text{sgn}(\ell)\theta_v, \quad H_n(\ell v) = \ell^{-n-2}H_n(v), \quad K_n(\ell v) = \ell^{-n-1}K_n(v), \\ c_{\ell v} &= \ell^2 c_v, \quad d_{\ell v} = \ell d_v, \quad c_{3,\ell v} = \ell^3 c_{3,v}, \quad d_{2,\ell v} = \ell^2 d_{2,v}, \quad e_{1,\ell v} = \ell e_{1,v}. \end{aligned} \quad (\text{RPD22})$$

The constants e, f stay fixed. This proves resonance invariance under every nonzero real scaling, including the signs of H_n, K_n . An inverse pole of exponent p has its norm constant multiplied by $|\ell|^{-p}$, exactly as follows also from the literal reparameterization $p_* + \tau\ell v = p_* + (\ell\tau)v$.

These formulas also classify repetitions across all cutoffs. An unoriented direction has a unique image argument $\phi \in [0, \pi)$ for the line $\mathbb{R}(A_v/B)$. It is resonant for some cutoff precisely when $\phi/\pi = m/q$ is rational with $0 < m < q$ and $\gcd(m, q) = 1$; its resonant cutoffs are exactly $D = jq - 1$, $j \geq 1$. The line $\phi = 0$ and every irrational argument are nonresonant at every cutoff. Two cutoffs D, E share exactly $\gcd(D + 1, E + 1) - 1$ resonance lines: the common fractions strictly between zero and one are the multiples of the reciprocal of that greatest common divisor. This follows by reducing the fractions in RPD20; it is an exact classification, not an interval test for an infinite list of equalities.

1.6 Curvature correction and the complete resonance transition

Let a real path have $p(\tau) = p_* + \tau v + \frac{1}{2}\tau^2 a + O(\tau^3)$, $v \neq 0$. Twice differentiating $G(\xi, p(\tau))$ shows that RPD9–14 still hold with

$$C_v \mapsto C_v + \Lambda(a), \quad K_n^{\text{path}} = K_n(v) + \frac{1}{2}DH_n(v)[a]. \quad (\text{RPD23})$$

Here H_v, F, B are unchanged to the orders in question. To verify the second equality directly, the increments are $\delta c_3 = \Re(\Lambda(a)\overline{A_v})$ and $\delta d_2 = -i\Re(B\overline{\Lambda(a)})$, with no increment in e_1, f . They equal $\frac{1}{2}Dc_v[a]$ and $\frac{1}{2}Dd_v[a]$. Differentiating q_v^{-1} in RPD13 therefore gives exactly the additional half derivative. The full resonant block uses this corrected upper-right coefficient; a tangent direction alone does not specify that coefficient.

We now prove a uniform transition formula rather than apply a fixed nonresonant asymptotic at changing directions. Fix any resonant nonzero v for cutoff D and any real w transverse to its line. The derivative $\kappa_D = DH_D(v)[w]$ is nonzero. To see this, use the exact transverse vector $w_0 = \Lambda^{-1}(iA_v)$. It has $Dc_v[w_0] = 0$ and $D\theta_v[w_0] = -\Im(A_v\bar{B})/(|A_v||B|) \neq 0$ at each interior root in RPD15. Those polynomial roots are simple, so $DH_D(v)[w_0] \neq 0$. Homogeneity in RPD22 gives $DH_D(v)[v] = -(D+2)H_D(v) = 0$. Every real w is a unique real combination of v, w_0 because their images are A_v, iA_v ; transversality means its w_0 coefficient is nonzero. This proves the assertion and also gives the exact derivative

$$DH_D(v)[w_0] = \frac{(i\lambda_v)^D}{c_v} U'_D(\theta_v) \frac{-\Im(A_v\bar{B})}{|A_v||B|}. \quad (\text{RPD24})$$

For real $\tau > 0$ and $\epsilon \rightarrow 0$, write $v_\epsilon = v + \epsilon w$ and evaluate the literal point $p_* + \tau v_\epsilon$. On a fixed compact neighborhood of v , c_{v_ϵ} is bounded away from zero. The complete analytic coefficient expansion proving RPD14 has uniformly bounded remainders there. It follows entry by entry that

$$\tau^{D+1}B_{D,s}(p_* + \tau v_\epsilon)^{-1} = \iota \begin{pmatrix} H_{D-1}(v_\epsilon)\rho_{D-1}/\rho_0 & \{H_D(v_\epsilon)/\tau + K_D(v_\epsilon)\}\rho_D/\rho_0 \\ 0 & H_{D-1}(v_\epsilon)\rho_D/\rho_1 \end{pmatrix} + O(\tau). \quad (\text{RPD25})$$

Here ι inserts the 2×2 corner at rows 0, 1 and columns $D-1, D$. For $D=1$ it is the identity insertion. The error is in the original matrix norm and is uniform for these v_ϵ . The two diagonal corner errors are $O(\tau)$ by RPD14 at degree $D-1$; the top-right error is $O(\tau)$ at degree D ; every other entry has degree at most $D-2$ and hence is $O(\tau)$ or smaller. This proves the formula, including its uniformity and all weights.

If $\epsilon(\tau)/\tau \rightarrow L \in \mathbb{R}$, the two divergent inverse singular values both have pole $D+1$, with constants the two singular values of

$$N_{D,s}(v; L, w) = \begin{pmatrix} H_{D-1}(v)\rho_{D-1}/\rho_0 & \{K_D(v) + L\kappa_D\}\rho_D/\rho_0 \\ 0 & H_{D-1}(v)\rho_D/\rho_1 \end{pmatrix}. \quad (\text{RPD26})$$

Indeed $H_D(v + \epsilon w) = \kappa_D \epsilon + O(\epsilon^2)$, so RPD25 converges to the displayed invertible block. Its determinant modulus is $L_{2,D,s}(v)$ by RPD15–16, independent of L . All remaining inverse singular values have the same finite positive limits as before. If instead $|\epsilon(\tau)|/\tau \rightarrow \infty$ while $\epsilon(\tau) \rightarrow 0$, RPD25's single growing entry and the same determinant identity give

$$\begin{aligned} \sigma_1(B_{D,s}^{-1}) &\sim |\kappa_D| \frac{\rho_D}{\rho_0} |\epsilon(\tau)| \tau^{-D-2}, \\ \sigma_2(B_{D,s}^{-1}) &\sim \frac{L_{2,D,s}(v)\rho_0}{|\kappa_D|\rho_D} |\epsilon(\tau)|^{-1} \tau^{-D}. \end{aligned} \quad (\text{RPD27})$$

For every $2 \leq r \leq N$ the exterior asymptotic is always $L_{r,D,s}(v)\tau^{-2N}$, since the direction tends to v and $g_0/\tau^2 \rightarrow c_v$. Thus no divergent crossover coefficient is hidden inside a falsely uniform fixed-direction estimate.

In particular take $\epsilon(\tau) = q_0 \tau^q$, $q_0 \neq 0$, $q > 0$. For $0 < q < 1$ the two pole exponents are $(D+2-q, D+q)$, with constants $|q_0 \kappa_D| \rho_D / \rho_0$ and its quotient into $L_{2,D,s}(v)$. These paths are differentiable at zero, but are not asserted to be twice differentiable. For $q = 1$ both poles are $D+1$ and RPD26 uses $L = q_0$; the path has acceleration $2q_0 w$, so the result agrees exactly with RPD23. For $q > 1$ both poles remain $D+1$ and the block is $N_{D,s}(v; 0, w)$.

For the original degree-three equation, the fully explicit specialization retains $b = |B|^2$, $c = c_v$, $R = R_v$:

$$\begin{aligned} H_3(v) &= iR(2cb - R^2)/c^4, \\ DH_3(v)[w] &= \frac{i\{(2cb - 3R^2)DR[w] + 2bR Dc[w]\}}{c^4} \quad \text{on } H_3(v) = 0, \\ Dc[w] &= 2\Re(\Lambda(w)\overline{A_v}), \quad DR[w] = 2\Re(\Lambda(w)\bar{B}), \\ H_2(v) &= (d_v^2 - c_v e)/c_v^3, \\ K_3(v) &= \frac{2c_v d_v e_{1,v} + 2c_{3,v} d_v e + (2c_v e - 3d_v^2) d_{2,v} - c_v^2 f}{c_v^4} \quad \text{on } H_3(v) = 0. \end{aligned} \tag{RPD28}$$

The three unoriented lines have $\theta_v = 0$ or $\theta_v^2 = 1/2$, with representatives RPD20 at $\phi = \pi/4, \pi/2, 3\pi/4$. The two singular poles are $(5, 3)$ off these lines and $(4, 4)$ on straight approaches along them. The full transition gives $(5 - q, 3 + q)$ for $0 < q < 1$ and $(4, 4)$ for $q \geq 1$; all exterior ranks 2, 3, 4 have pole eight with constants RPD16. RPD28 follows by differentiating its displayed H_3 at a zero and by extracting the next reciprocal coefficient; it includes the -4 center terms through RPD9.

1.7 Exact return to the positive original period and actual unit

The period inversion is an exact coordinate map, not a constant rescaling of a curved two-parameter path:

$$(z, x) \mapsto (u, x) = (1/z, x), \quad (v_z, v_x) \mapsto (-u_*^2 v_z, v_x), \quad da(v) = i\alpha v_x. \tag{RPD29}$$

Its differential is invertible because $z_* > 0$. Thus every resonance vector in original (u, x) coordinates is $(-u_*^2 v_z, v_x)$ with v given in RPD20, and the actual unit is always $\alpha(1 + ix)$. For a straight original-period path $(u, x) = (u_*, x_*) + \tau(\nu_u, \nu_x)$, the exact expansion is

$$v = (-z_*^2 \nu_u, \nu_x), \quad a = (2z_*^3 \nu_u^2, 0), \quad K_D^{(u,x)} = K_D(v) + \frac{1}{2} DH_D(v)[a]. \tag{RPD30}$$

This follows by differentiating $z = 1/u$ twice and applying RPD23. The directional zeroth coefficient is c_v , all weights remain RPD11, and the resonant leading block uses the full RPD30 term. When only the period varies with $\nu_u = 1$, this reproduces the original $|u_*|^{2p}$ factor at pole exponent p . For arbitrary nonzero ν_u there is also the factor $|\nu_u|^{-p}$. At resonance the curvature vector is parallel to v and its derivative term vanishes by RPD22. For a mixed direction there is no general license to drop that term. The isolated real singular point, its exact resonance lines, and every crossover above concern the full geometric family defined in RPD1. No step puts an arithmetic zeta quartet on this family's singular point or supplies that unconstructed arithmetic map.

1.8 Certified original-coordinate constants for the directional geometry

The derivative isomorphism in RPD3–4 can also be enclosed without evaluating a new truncated series. Use the complete RPZ certificate's exact dyadic matrix P and its enclosing matrix $E = I - PJ$. Since P is exactly invertible, the identity $J = P^{-1}(I - E)$ encloses the full original real Jacobian, including all the tails already proved in RPZ5–14. At the exact certified root,

$f_1 = 0$, so the product rule gives $G_x = f_2(J_{01} + iJ_{11})$. Combine this with the saved full enclosures for G_z and G_ξ . Outward interval multiplication, determinants and inverses give

$$\begin{aligned} G_x &\in [-0.19190307 \pm 6.75 \cdot 10^{-9}] + i[0.83428595 \pm 6.63 \cdot 10^{-9}], \\ \Delta &\in [-88.12638 \pm 6.25 \cdot 10^{-6}], \\ \mathcal{Q} &\in \begin{pmatrix} [82176.45 \pm 5.29 \cdot 10^{-3}] & [-229.0362 \pm 5.50 \cdot 10^{-5}] \\ [-229.0362 \pm 5.50 \cdot 10^{-5}] & [0.7328598 \pm 4.42 \cdot 10^{-8}] \end{pmatrix}, \\ \det \mathcal{Q} &\in [7766.26 \pm 7.77 \cdot 10^{-3}]. \end{aligned} \tag{RPD31}$$

Here $[m \pm r]$ denotes the enclosing real interval, not a decimal substitution for the derivative. In particular these intervals independently prove negative orientation and positive definiteness. The complete propagation code is `independent/certify_real_pair_direction_constants.py`; its receipt is `independent/REAL_PAIR_DIRECTION_CONSTANTS.json`. The receipt binds the complete root certificate by SHA-256:

735c347ad1afe823bdf8cae183bcdcc614bfcdce49f13fcf79829e6b8a7769c.

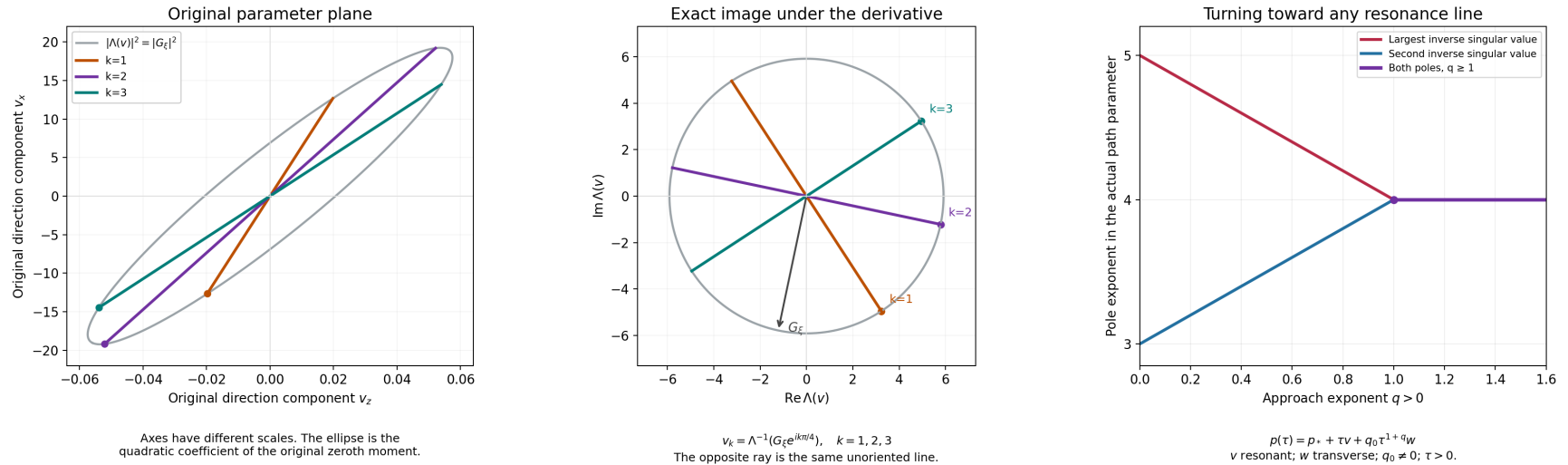
It supplies outward enclosures of all coordinates of the three exact degree-three vectors

$$\Lambda^{-1}(Be^{ik\pi/4}), \quad k = 1, 2, 3.$$

Their nonzero lengths are retained; they are representatives of the full $r > 0$ families in RPD20. The figure draws their interval midpoints only to display the tangent geometry. Their exact definition, every resonance line and all claimed exponents follow from RPD4 and RPD15–28, rather than from the drawing. Interval arithmetic is Johansson’s Arb [7]; the original maps and metrics remain [1, 6].

The same real singular point: exact resonance directions and the pole transition

$$\Lambda(v) = G_z v_z + G_x v_x, \quad \det \Lambda \in [-88.12638 \pm 6.25 \times 10^{-6}] \neq 0$$



Degree $D=3$: other straight directions have poles $(5, 3)$; each displayed resonance line has $(4, 4)$.
Every inverse exterior rank 2, 3 and 4 has pole 8 throughout the transition, with the full original Gamma constants.

Proof: RPD3–5, RPD20–28 and RPD31. Tangent geometry uses certified interval midpoints; it is not a plot of a finite zero curve.
Original period and metric: PCL1–16, WCF1–6 (pinned sources in the proof). Inclusion arithmetic: Fredrik Johansson, Arb, arXiv:1611.02831v1.

Figure 1: RPD3–5 gives the exact original real derivative and its complete quadratic form; RPD20–22 constructs every resonance line. Three degree-three representatives and their antipodal rays are displayed at the midpoints of the enclosing intervals in RPD31. The left axes have different scales; the middle panel is the image under the exact derivative, not an asserted isometry. RPD23–30 proves the complete curved-path pole transition and original-period correction. The plot records tangent geometry, not a finite spectral zero curve. Full original family and metric: [1, 6]; inclusion arithmetic: Fredrik Johansson [7].

References

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Pinned public proof: PCL1–16.
- [2] Split-Zero programme, result SZ-20260921-005, `REAL_PAIR_ALL_CUTOFFS.tex`, RPAC1–15.
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- [3] Split-Zero programme, result SZ-20260921-005, `REAL_PAIR_COLLISION.tex`, RPCJ1–23.
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- [4] Split-Zero programme, result SZ-20260921-005, `REAL_PAIR_ES_RETURN.tex`, RPE1–26.
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- [5] Split-Zero programme, result SZ-20260921-005, `REAL_PAIR_ZERO_BODY.tex`, RPZ1–19.
Pinned complete public proof: RPZ1–19.
- [6] Split-Zero programme, *A polynomial forward bound for the full weighted conductor*, `WEIGHTED_CONDUCTOR_FORWARD.tex`, WCF1–6, source intake 17 September 2026. This source credits the Meixner–Pollaczek dictionary to the NIST DLMF Chapter 18 by T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, equations 18.19.8–9, 18.22.8 and 18.23.7; its original phases and masses are retained here, and no new orthogonality theorem is asserted.
Pinned public proof: WCF1–6.
- [7] Fredrik Johansson, *Arb: Efficient Arbitrary-Precision Midpoint-Radius Interval Arithmetic*, original author source, arXiv:1611.02831v1 (9 November 2016), <https://arxiv.org/abs/1611.02831v1>. Source sections “Features and example applications”, “Radii and magnitude bounds” and “Precision and bounds” support the stated inclusion arithmetic. The complete FPZ series-tail and zero-count proofs are given here; they are not attributed to the software paper.